2023 Summer Qual

III.

Let G_1 and G_2 s.t. $AG_1A^TA = A$,
$AG_2A^TA = A$

$$0 = (AG_1A^TA - AG_2A^TA)(G_1^TA^T - G_2^TA^T)$$
$$= \underbrace{(AG_1A^T - AG_2A)}_{=B}(AG_1A^T - AG_2A^T)$$

$$= B^2$$

$$\Rightarrow B = 0$$

$$\Rightarrow AG_1A^T - AG_2A = 0 \rightsquigarrow AG_1A^T = AG_2A^T$$

So, $A(A^TA)^-A^T$ is invariant to the choice of $(A^TA)^-$. ∎

$\boxed{2}$

$$\mathrm{mgf}_{y^TAy}(t) = \mathbb{E}\left[e^{t y^T A y}\right]$$

$$= \int_{-\infty}^{\infty} \frac{1}{(2\pi)^p |\Sigma|^{1/2}} e^{t y^T A y} e^{-\frac{1}{2}(y-\mu)^T \Sigma^{-1}(y-\mu)} dy$$

$$= \int_{-\infty}^{\infty} \frac{1}{(2\pi)^p |\Sigma|^{1/2}} e^{-\frac{1}{2}\left[y^T(I-2tA\Sigma)\Sigma^{-1}y - 2y^T\Sigma^{-1}\mu + \mu^T\Sigma^{-1}\mu\right]} dy$$

For small t, $\exists\, (I-2tA\Sigma)^{-1}$

$$\theta^T = \mu^T(I-2tA\Sigma)^{-1}, \quad V^{-1} = (I-2tA\Sigma)\Sigma^{-1}$$

$$\mathrm{mgf}_{y^TAy}(t) = \int_{-\infty}^{\infty} \frac{1}{(2\pi)^p |\Sigma|^{1/2}} e^{-\frac{1}{2}(y-\theta)^T V^{-1}(y-\theta)} \cdot e^{-\frac{1}{2}\left[\mu^T\Sigma^{-1}\mu - \theta^T V^{-1}\theta\right]} dy$$

$$= |I-2tA\Sigma|^{-1/2} e^{-\frac{1}{2}\underbrace{\mu^T\left[\Sigma^{-1}(I-(I-2t\Sigma A)^{-1})\right]\mu}_{\text{scalar}}}$$

$$= |I-2tA\Sigma|^{-1/2} \exp\left\{-\frac{1}{2}\mu^T\left[(I-(I-2tA\Sigma)^{-1})\Sigma^{-1}\right]\mu\right\}$$

$\blacksquare$

$\boxed{3}$.

① $c^T\hat{\beta} = c^T(X^TX)^{-1}X^TY$ is a linear combination of Y

② $E[c^T\hat{\beta}] = c^T\beta$: unbiased.

③ Let $\tilde{\beta} = d^TY$ be any linear unbiased estimator of $c^T\beta$

$$E[\tilde{\beta}] = d^TX\beta = c^T\beta \Rightarrow d^TX = c^T$$

$$0 \le var(\tilde{\beta} - c^T\hat{\beta}) = var((d^T - c^T(X^TX)^{-1}X^T)Y)$$

$$= (d^T - c^T(X^TX)^{-1}X^T)\sigma^2 I(d - X(X^TX)^{-1}c)$$

$$= \sigma^2(d^Td - c^T(X^TX)^{-1}c - c^T(X^TX)^{-1}c + c^T(X^TX)^{-1}c)$$

$$= var(\tilde{\beta}) - var(c^T\hat{\beta})$$

$\Rightarrow var(c^T\hat{\beta}) \le var(\tilde{\beta})$: minimum variance.

$\therefore c^T\hat{\beta}$ is BLUE of $c^T\beta$ $\boxtimes$

四.

(a) $Y = X_1\beta_1 + \varepsilon$ RM

$Y = X_1\beta_1 + X_2\beta_2 + \varepsilon$ FM

$$R^2_{RM} = \frac{Y^T(H_{X_1} - H_1)Y}{Y^T(I - H_1)Y}, \quad R^2_{FM} = \frac{Y^T(H_X - H_1)Y}{Y^T(I - H_1)Y}$$

$$H_X = H_{X_1} + H_{X_{2\perp}}, \quad X_{2\perp} = (I - H_{X_1})X_2$$

$$X = (X_1, X_{2\perp})$$

$$X(X^TX)^{\dagger}X^T = (X_1 \quad X_{2\perp}) \begin{pmatrix} X_1^TX_1 & 0 \\ 0 & X_{2\perp}^TX_{2\perp} \end{pmatrix} \begin{pmatrix} X_1^T \\ X_{2\perp}^T \end{pmatrix}$$

$$= H_{X_1} + H_{X_{2\perp}}$$

$$R^2_{FM} - R^2_{RM} \Leftrightarrow Y^T(H_X - H_1)Y - Y^T(H_{X_1} - H_1)Y$$

$$= Y^T(H_X - H_{X_1})Y$$

$$= Y^T \underbrace{H_{X_{2\perp}}}_{\substack{\text{symm \& idemp} \\ \Rightarrow \text{p.s.d}}} Y \geq 0$$

$\blacksquare$

(b) $X_* = XA$, A is full rank.

$$H_{X_*} = X_*(X_*^TX_*)^{-1}X_*^T = XA(A^TX^TXA)^{-1}A^TX$$

$$= XAA^{-1}(X^TX)^{-1}(A^T)^{-1}A^TX$$

$$= X(X^TX)^{-1}X^T = H_X$$

$$\frac{Y^T(H_X - H_1)Y}{Y^T(I - H_1)Y} = \frac{Y^T(H_X - H_1)Y}{Y^T(I - H_1)Y} = R^2$$

$$Z = CY$$

$$\frac{Z^T(H_X - H_1)Z}{Z^T(I - H_1)Z} = \frac{\cancel{c^2}\, Y^T(H_X - H_1)Y}{\cancel{c^2}\, Y^T(I - H_1)Y} = R^2$$

$\boxtimes$

(c) $$R^2 = \frac{SSR}{SST} = \frac{SSR}{SSR + SSE}$$ & $$(H_X - H_1)(I - H_X) = H_X - H_X - H_1 + H_1 = 0$$

$$= \frac{\frac{SSR}{\sigma^2}}{\frac{SSR}{\sigma^2} + \frac{SSE}{\sigma^2}} = \frac{\chi^2(p)}{\chi^2(p) + \chi^2(n-p-1)}$$

$$= \frac{\Gamma(\frac{p}{2}, 2)}{\Gamma(\frac{p}{2}, 2) + \Gamma(\frac{n-p-1}{2}, 2)} \sim \text{Beta}\left(\frac{p}{2}, \frac{n-p-1}{2}\right)$$

$\boxtimes$